Exception handling programs

Null pointer exception

Steps
Start the program.
Create a String variable name and assign null.
Enter the try block.
Try to find the length of name using name.length().
Since name is null, a NullPointerException occurs.
The catch block catches the exception.
Display the message "we get an nullpointer exception" along with the exception message.
End the program.

Algorithm
Step 1: Start.
Step 2: Declare a String variable name and initialize it to null.
Step 3: Begin the try block.
Step 4: Execute name.length().
Step 5: If name is null, a NullPointerException is generated.
Step 6: Catch the exception using catch (Exception e).
Step 7: Display the exception message.
Step 8: Stop.

Code and output

```
// null ponter exception

public class EphProm1 {
public static void main(String[] args) {
String name  = null;
try {
System.out.println(name.length());
} catch (Exception e) {
// TODO: handle exception
System.out.println("we get an nullpointer exception" + e.getMessage());
}
}
}
```

```
null
PS C:\Users\bsai8\OneDrive\Desktop\javaprograms> javac EphProm1.java
PS C:\Users\bsai8\OneDrive\Desktop\javaprograms> java EphProm1
we get an nullpointer exceptionCannot invoke "String.length()" because "<local1>" is null
```

Array index out of bounds exception

Steps
Start the program.
Create an integer array a with 5 elements.
Enter the try block.
Try to access a[6].
Since valid indexes are only 0 to 4, an ArrayIndexOutOfBoundsException occurs.
The catch block catches the exception.
Display the exception message using aioobe.getMessage().
The finally block executes regardless of whether an exception occurs.
Display the message from the finally block.
End the program.

Algorithm
Step 1: Start.
Step 2: Declare and initialize an integer array a with 5 elements.
Step 3: Begin the try block.
Step 4: Access the element at index 6.
Step 5: Check whether index 6 is within the array range.
Step 6: Since index 6 is invalid, generate ArrayIndexOutOfBoundsException.
Step 7: Catch the exception using catch (ArrayIndexOutOfBoundsException aioobe).
Step 8: Display the exception message.
Step 9: Execute the finally block and display its message.
Step 10: Stop

Code and output

```
public class EphProm2 {
public static void main(String[] args) {
int a[] = {10,20,30,40,50};
try {
System.out.println(a[6]);
} catch (ArrayIndexOutOfBoundsException aioobe) {
// TODO: handle exception

System.out.println("i caught an exception named called aioobe" + aioobe.getMessage());
}
finally{
System.out.println("we sucessfully the eception");
}
}
}
```

```
PS C:\Users\bsai8\OneDrive\Desktop\javaprograms> javac EphProm2.java
PS C:\Users\bsai8\OneDrive\Desktop\javaprograms> java EphProm2
i caught an exception named called aioobeIndex 6 out of bounds for length 5
we sucessfully the eception
```

String index out of bounds exception

Steps
Start the program.
Create a String variable man and assign "hemanth".
Enter the try block.
Try to access the character at index 9 using man.charAt(9).
The string "hemanth" has only 7 characters, so valid indexes are 0–6.
Since index 9 is invalid, a StringIndexOutOfBoundsException occurs.
The catch block catches the exception.
Display the exception message using sioobe.getMessage().
The finally block executes and displays its message.
End the program.

Algorithm
Step 1: Start.
Step 2: Declare a String variable man and initialize it with "hemanth".
Step 3: Begin the try block.
Step 4: Access the character at index 9.
Step 5: Check whether index 9 is valid.
Step 6: Since index 9 is outside the string range, generate StringIndexOutOfBoundsException.
Step 7: Catch the exception using catch (StringIndexOutOfBoundsException sioobe).
Step 8: Display the exception message.
Step 9: Execute the finally block and display its message.
Step 10: Stop.

Code and output

```
public class EphProm3 {
public static void main(String[] args) {
String man = "hemanth";
try {
System.out.println(man.charAt(9));
} catch (StringIndexOutOfBoundsException sioobe) {
// TODO: handle exception
System.out.println("we caught an exception called sioobe" + sioobe.getMessage());

}
finally{
System.out.println("we get sioobe exception");
}
}

}
```

```
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PS C:\Users\bsai8\OneDrive\Desktop\javaprograms> javac EphProm3.java
PS C:\Users\bsai8\OneDrive\Desktop\javaprograms> java EphProm3
we caught an exception called sioobeIndex 9 out of bounds for length 7
we get sioobe exception
```

Class cast exception

Steps
Start the program.
Create an Object variable obj.
Store a String object "hemanth" in obj.
Enter the try block.
Try to convert the String object into an Integer.
Since a String cannot be cast to Integer, a ClassCastException occurs.
The catch block catches the exception.
Display the exception message using cce.getMessage().
Stop the program.
Algorithm
Step 1: Start.
Step 2: Create an Object variable obj and assign a String object to it.
Step 3: Begin the try block.
Step 4: Cast the Object into an Integer.
Step 5: Check whether the object can be converted to Integer.
Step 6: Since the actual object is a String, generate ClassCastException.
Step 7: Catch the exception using catch (ClassCastException cce).
Step 8: Display the exception mess
age.
Step 9: Stop.

Code and output

```
// classcast exception
public class EphProm4 {
public static void main(String[] args) {
object obj = new String("hemanth");
try {
Integer number = (Integer) obj;
} catch (ClassCastException cce) {
// TODO: handle exception
System.out.println("we catch an exception called cce"+ cce.getMessage());

}
}
}
```

```
Caught ClassCastException: class java.lang.String cannot be
cast to class java.lang.Integer (java.lang.String and java.l
ang.Integer are in module java.base of loader 'bootstrap')

[Program finished]
```

Number format exception

Steps
Start the program.
Create a String variable val and assign "hemu0912".
Enter the try block.
Use Integer.parseInt(val) to convert the String into an integer.
Since "hemu0912" contains letters, it cannot be converted into an integer.
A NumberFormatException occurs.
The catch block catches the exception.
Display the exception message using nfe.getMessage().
Stop the program.

Algorithm
Step 1: Start.
Step 2: Declare a String variable val and initialize it with "hemu0912".
Step 3: Begin the try block.
Step 4: Convert val into an integer using Integer.parseInt().
Step 5: Check whether the String contains a valid integer value.
Step 6: Since "hemu0912" contains non-numeric characters, generate NumberFormatException.
Step 7: Catch the exception using catch (NumberFormatException nfe).
Step 8: Display the exception message.
Step 9: Stop.

Code and output

```
public class EphProm5 {
public static void main(String[] args) {
String val = "hemu0912";
try {
int number = Integer.parseInt(val);
System.out.println();
} catch (NumberFormatException nfe) {
// TODO: handle exception
System.out.println("we catch an nfe" + nfe.getMessage());
}
}
}
```

```
PS C:\Users\bsai8\OneDrive\Desktop\javaprograms> javac EphProm5.java
PS C:\Users\bsai8\OneDrive\Desktop\javaprograms> java EphProm5
we catch an nfeFor input string: "hemu0912"
PS C:\Users\bsai8\OneDrive\Desktop\javaprograms>
```

6. Illegal argument exception

Steps
Start the program.
Define the check(int age) method.
Pass -2 as the age to the check() method.
Check whether age < 0.
Since -2 is negative, throw IllegalArgumentException.
The catch block catches the exception.
Display the exception message using iae.getMessage().
The finally block executes.
Display the final message.
Stop the program.

Algorithm
Step 1: Start.
Step 2: Define the check() method with an integer parameter age.
Step 3: Check whether age is less than 0.
Step 4: If age < 0, throw IllegalArgumentException.
Step 5: Call check(-2) inside the try block.
Step 6: Catch the exception using catch (IllegalArgumentException iae).
Step 7: Display the exception message.
Step 8: Execute the finally block.
Step 9: Display "we successfully cathced the exception".
Step 10: Stop.

Code and output

```
public class EphProm6 {
static void check(int age){
if(age < 0){
throw new IllegalArgumentException("age cannot be negative" + age);
}
System.out.println("age set to " + age);
}

public static void main(String[] args) {

try {
check(-2);
} catch (IllegalArgumentException iae) {
// TODO: handle exception
System.out.println("we catch iae " + iae.getMessage());
}
finally{
System.out.println("we successfully cathced the exception");
}
}
}
```

```
PS C:\Users\bsai8\OneDrive\Desktop\javaprograms> javac EphProm6.java
PS C:\Users\bsai8\OneDrive\Desktop\javaprograms> java EphProm6
we catch iae age cannot be negative-2
we successfully cathced the exception
```

7.negative array size exception

Steps
Start the program.
Enter the try block.
Create an integer variable max and assign -5.
Try to create an integer array with size -5.
Since an array size cannot be negative, NegativeArraySizeException occurs.
The catch block catches the exception.
Display the exception message using nase.getMessage().
Stop the program.

Algorithm
Step 1: Start.
Step 2: Begin the try block.
Step 3: Initialize max = -5.
Step 4: Create an integer array with size max.
Step 5: Check whether the array size is negative.
Step 6: Since max is -5, generate NegativeArraySizeException.
Step 7: Catch the exception using catch (NegativeArraySizeException nase).
Step 8: Display the exception message.
Step 9: Stop.

Code and output

```
// negative arraysize exception
public class EphProm7 {
public static void main(String[] args) {
try {
int max = -5;
int b[] = new int[max];
} catch (NegativeArraySizeException nase) {

// TODO: handle exception
System.out.println("catcehd nase" + nase.getMessage());
}
}
}
```

```
PS C:\Users\bsai8\OneDrive\Desktop\javaprograms> javac EphProm7.java
PS C:\Users\bsai8\OneDrive\Desktop\javaprograms> java EphProm7
catcehd nase-5
```

8.array store exception

Steps
Start the program.
Create an Object array a, but the actual array is a String array of size 3.
Enter the try block.
Try to store an Integer value 9 into the String array.
Since the array can store only String objects, an ArrayStoreException occurs.
The catch block catches the exception.
Display the exception message using ase.getMessage().
The finally block executes.
Display "the process is doned".
Stop the program.

Algorithm
Step 1: Start.
Step 2: Create an Object array that actually refers to a String array of size 3.
Step 3: Begin the try block.
Step 4: Try to store an Integer value into the array.
Step 5: Check whether the value is compatible with the actual array type.
Step 6: Since Integer is not a String, generate ArrayStoreException.
Step 7: Catch the exception using catch (ArrayStoreException ase).
Step 8: Display the exception message.
Step 9: Execute the finally block.
Step 10: Display "the process is doned".
Step 11: Stop.

Code and output

```
// arraystoreexception
public class EphProm8 {
public static void main(String[] args) {

Object a[] = new String[3];
try {
 a[0] = Integer.valueOf(9);
} catch (ArrayStoreException ase) {
 // TODO: handle exception
System.out.println("catched is ase"  +ase.getMessage());

}
finally{
System.out.println("the process is doned");
}
}
}
```

```
PS C:\Users\bsai8\OneDrive\Desktop\javaprograms> javac EphProm8.java
PS C:\Users\bsai8\OneDrive\Desktop\javaprograms> java EphProm8
catched is asejava.lang.Integer
the process is doned
```

9.TNPE

Steps
Start the program.
Enter the try block.
Explicitly throw a TypeNotPresentException.
Pass the error message and a ClassNotFoundException as the cause.
The TypeNotPresentException is generated.
The catch block catches it using tnpe.
Display the exception message using tnpe.getMessage().
The finally block executes.
Display "catched done".
Stop the program.

Algorithm
Step 1: Start.
Step 2: Begin the try block.
Step 3: Create and throw a TypeNotPresentException with a message and cause.
Step 4: Check whether the thrown exception is TypeNotPresentException.
Step 5: Catch the exception using catch (TypeNotPresentException tnpe).
Step 6: Display the exception message using tnpe.getMessage().
Step 7: Execute the finally block.
Step 8: Display "catched done".
Step 9: Stop.

Code and output

```
// type not prresentexception
public class EphProm9 {
public static void main(String[] args) {
try {
throw new TypeNotPresentException("missing class of not same type", new ClassNotFoundException());
} catch (TypeNotPresentException tnpe) {
// TODO: handle exception
System.out.println("the exception is "  + tnpe.getMessage());

}
finally{
System.out.println("catched done");
}
}
}
```

```
PS C:\Users\bsai8\OneDrive\Desktop\javaprograms> javac EphProm9.java
PS C:\Users\bsai8\OneDrive\Desktop\javaprograms> java EphProm9
the exception is Type missing class of not same type not present
catched done
```

10.Illegal state exception

Steps
Start the program.
Create a static boolean variable start and initialize it to false.
Create the post() method.
Inside post(), check whether start is false.
If start is false, throw IllegalStateException.
In the main() method, call post() inside the try block.
The exception occurs because start is false.
The catch block catches the IllegalStateException.
Display the error message using ise.getMessage().
The finally block executes.
Display "the exception is done".
Stop the program.

Algorithm
Step 1: Start.
Step 2: Initialize start = false.
Step 3: Define the post() method.
Step 4: Check whether start is false.
Step 5: If start is false, throw IllegalStateException.
Step 6: Call post() inside the try block.
Step 7: Catch the exception using catch (IllegalStateException ise).
Step 8: Display the exception message.
Step 9: Execute the finally block.
Step 10: Display the final message.
Step 11: Stop.

Code and output

```
// illegal state exception
public class EphProm10 {
static boolean start = false;
static void post(){
if(!start){
throw new IllegalStateException("cannot
start , post can't start to post");
}
System.out.println("post is ready to social
media ");
}
public static void main(String[] args) {
try {
System.out.println("the posting is stopped");
} catch (IllegalStateException ise) {
// TODO: handle exception
System.out.println("error" +
ise.getMessage());
}
finally{
System.out.println("the exception is done");
}



}
}
```

```
PS C:\Users\bsai8\OneDrive\Desktop\javaprograms> javac EphProm10.java
PS C:\Users\bsai8\OneDrive\Desktop\javaprograms> java EphProm10
the posting is stopped
the exception is done
```